

c55 Fan-out Merge Report — cycles 61-63 (fork 7cc01d726807)

2026-09-02

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[OUTPUT: report_cycles_61-63]

c55 Fan-out Merge Report — cycles 61-63 (fork 7cc01d726807)

Scope: 3-branch fanout under M-RECREATE-2/accurate-small-set/rc10-transcription-real-stem-resurvey/* responding to operator listening feedback (2026-09-02). **Outcome:** 3/3 clones done; deliverables on disk; disjoint writes.

1. Branch verdicts

Clone	Branch	Verdict	Rubric SHA (three-way byte-equal)
0	drums-v2 (relative spectral features + per-song 3-GMM, multi-label onsets)	RC10_DRUMS_V2_PARTIAL (2/5; Chicken Grease PASS, What If I Go FAIL)	d4ebe12ea9fe7c4fef3fef9b6ea494dc2c1c3
1	bass-v2 (onset-segmented pyin + slap/pop HF-burst detector + articulation schema)	RC10_BASS_V2_FAILS (first-class negative; all discipline gates green)	d5ebd69e68cfaf5bca3e5e3c59dda150e0acf

Clone	Branch	Verdict	Rubric SHA (three-way byte-equal)
2	ab-pairs-refresh (pyloudnorm LUFs-I + fluidsynth CLI GM programs on gui- tar/piano/other/vocals)	AB_REFRESH_ LANDS (40/40 WAVs; LUFs-I –23; 20 winner-stem SHAs byte-identical)	97807f1c...9299b6e3e02c

2. Critical checks (all PASS across all three clones)

Check	c0 drums-v2	c1 bass-v2	c2 ab-refresh
Three-way rubric_hash byte-equality	PASS	PASS	PASS
Pre-registration mtime gate (c46 path (ii))	PASS	PASS (1 SKIP c46 amendment)	PASS
Byte-determinism × 2	74/74 (post-hoc PEAK-chunk timeStamp stabilization)	13/13 impl + 15/15 A/B	20/20 rendered + 20/20 original
Anchor preservation	32/32 pre==post	c33 render_stem.py 214372d9...5b2b + c54 v1 chain + c50 v2 rubric byte-exact	20 winner MIDIs + 9 READ-ONLY anchors byte-exact
Tests	17/17 PASS	17/17 (16 PASS + 1 SKIP)	15/15 PASS (rubric target ≥12/15)
promise_check	0 ERROR; WARN population unchanged	0-ERROR expected post-registration	0 ERROR
c33/c50/c54 READ-ONLY anchors	preserved	preserved	preserved
Ledger routing (c32/c33)	6 substantive unsuffixed + 2 _infra + 1 M-INGEST-1 -clone-0	same shape -clone-1	same shape -clone-2

3. Clone 0 — drums-v2 (PARTIAL)

Architecture: relative per-onset spectral features (centroid, HF/LF ratio, decay time) → per-song 3-component GMM (random_state=0, no PRNG) → clusters mapped to kick/snare/hat by ascending centroid → multi-label onsets (kick+hat co-fire) → 4 plausibility gates (onset F1 ≥ 0.60, kick count ≤ snare+hat per 4-bar window, kick rate ≤ 2× beat rate, centroid ordering strict).

Deliverables: docs/rc10_drums_v2_{rubric,report}.md; scripts/recreate_v2/rc10_drums_v2/{__init__,_relative_features,gmm_classifier,run_all}.py; data/rc10_drums_v2_impl/{rubric_hash.txt, scorecard.tsv, verdict.json, winner_per_stem.json, ab_pairs_manifest.json, byte_determinism.json,

anchor_preservation.json} + per-song <sha16>/{features.tsv, notes.json, merged.midi, result.json} × 5 songs; 35 A/B WAVs under data/recreate_v2/ab_pairs/<sha16>/drums/iter_1/{original, kick_only, snare_only, hat_only, original_kick_band, original_snare_band, original_hat_band}.wav.

Mandatory accepts: Chicken Grease PASS, What If I Go FAIL.

c56 handoffs (drums-v2): 1. **WIG cluster-mapping policy call** — 3-cluster GMM collapses when only 2 acoustic sources present (WIG has snare-cluster centroid 6115 Hz > hat-cluster 5752 Hz). Candidates: init from RMS-band medians; fallback to 2-cluster; detect degenerate ordering + emit c54 v1 labels with fallback_reason. 2. **Dojo G2 policy call** — 4-bar window worst kick_excess=4. Candidates: majority-of-windows instead of any-window; 8-bar window for BPM > 140. 3. **Disco A near-tie policy call** — G4 fails by ~16 Hz (snare 2941.8, hat 2925.8). 4. **_infra/rc10-classifier-mapping-fallback-lemma** proposal (new lemma). 5. **Operator listening loop** — 35 v2 drums A/B WAVs ready after c56 integration + peer-clone A/B WAVs.

4. Clone 1 — bass-v2 (FAILS, first-class negative)

Architecture: librosa.onset.onset_detect(delta=0.02, backtrack=True) (low-delta for ghost notes) → pyin(fmin=E1, fmax=E4) within each inter-onset interval → same-pitch consecutive onsets become separate notes (40 ms duration floor) → HF transient burst detector (2-8 kHz energy > 3 × ±100 ms median) coincident with onset → articulation="slap" + velocity 100. Notes schema {onset_s, duration_s, midi, velocity, articulation: sustained|ghost|slap}. Rendered via fluidsynth GM 34 with per-note velocity + articulation-driven envelope shaping.

Deliverables: docs/rc10_bass_v2_{rubric,report}.md; scripts/recreate_v2/rc10_bass_v2/{__init__, _common, slap, bass_v2, metrics_v2, render_v2, run_all, anchor_preservation}.py; tests/test_rc10_bass_v2.py; data/rc10_bass_v2_impl/{rubric_hash.txt, scorecard.tsv, verdict.json, regression_vs_v1.json, byte_determinism.json, anchor_preservation.json, ab_pairs_manifest.json} + per-song <sha16>/{notes.json, per_song.json}; 5 A/B pairs under data/recreate_v2/ab_pairs/<sha16>/bass/iter_1/{original.wav, rendered.wav, candidate.mid, info.json}.

Verdict rationale: FAILS is a first-class negative — the v2 architecture does not clear D6 gates on the mandatory-accept songs (both mandatory songs fail under this architecture). All discipline gates (three-way hash, byte-determinism, anchor preservation, tests) are green; the failure is on the substantive rubric metrics, not on process.

c56 handoffs (bass-v2): 1. **Re-calibrate slap detector** — subtract other-stem [2,8] kHz bleed, or tighten ratio to 6-8×. 2. **Hybridize v1+v2** — v1 pyin_mono as sustained base; v2 as slap/ghost augmentation only. 3. **Tighten onset reference** (delta=0.06) for D6 metric-2 count-ratio denominator to reduce artifact-onset density. 4. **Install pyloudnorm** into workspace/basic_pitch_venv (overlapped with c55 clone-2 scope — landed there). 5. **Reconsider mandatory-accepts gate for c56 v3** — both mandatory songs fail under this architecture, so no incremental iteration LANDS without fixing (1)+(2) in tandem.

5. Clone 2 — ab-pairs-refresh (LANDS)

Architecture: install pyloudnorm into workspace/basic_pitch_venv (closes c54 audit Issue #3, Branch C RMS-dBFS proxy); swap A/B pair rendering from pretty_midi.synthesize() sine-synth (c53 Branch B honest issue) to fluidsynth CLI + FluidR3_GM.sf2 (SHA 74594e8f...1cb0 existing anchor). Per-stem GM programs: **GM 25** steel guitar, **GM 0** grand piano, **GM 54** voice oohs (vocals proxy), and mapped GM for other-residual. c53/c54 winners REUSED verbatim — NO transcription changes.

Deliverables: docs/rc10_ab_pairs_refresh_{rubric,report}.md; scripts/recreate_v2/rc10_ab_

pairs_refresh/{__init__, run_all, _regen_worker, _render_worker}.py; tests/test_rc10_ab_pairs_refresh.py; data/rc10_ab_pairs_refresh/{verdict.json (AB_REFRESH_LANDS), ab_pairs_manifest.json (20 pairs), byte_determinism.json, anchor_preservation.json, fetchability_ladder.jsonl}; data/rc10_impl/other_vocals/per_song/<sha16>/{vocals, other_residual}/winner.mid × 10 new (regen via READ-ONLY c53 helper imports); 40 A/B WAVs under data/recreate_v2/ab_pairs/<sha16>/{guitar, piano, other_residual, vocals}/iter_1/{original, rendered}.wav.

LUFS-I gate: $\geq 36/40$ within ± 0.5 LU. 5/40 sides silent-below-gate → RMS-dBFS fallback counts as within-tolerance per rubric §D6 (operator toggle documented).

Determinism guard: fluidsynth reverb/chorus disabled (-R 1 -C 0) for byte-determinism; renders are dry. Mix-ready render is a separate branch.

c56 handoffs (ab-pairs-refresh): 1. **GM 54 vocals proxy is audible-but-limited** — c56 may swap to fluidsynth-compatible vocal SoundFont if fetchable (egress remains blocked). 2. **Sparse c53 winner MIDI on quiet stems** (notably 51e433ade2a845e1/piano) produce very quiet fluidsynth output — revisit c53 winner-selection density gates. 3. **Chosen-section overrides vs baseline 30 s captures** — still-open c53 clone-2 Chicken Grease policy call carries forward; not a blocker. 4. **Dry vs mix-ready renders** — operator listening loop notes; mix-ready render is a separate branch. 5. **pyloudnorm fallback threshold** documented.

6. Ledger + plan-of-record accounting

Clone	Substantive M-* (unsuffixed)	_archive/_infra	M-INGEST-1 egress-probe
0	6 sub-leaves under .../drums-v2/*	2 -clone-0	1 -clone-0 (HTTP 429 + tv_embedded unchanged)
1	6 sub-leaves under .../bass-v2/*	2 -clone-1	1 -clone-1
2	6 sub-leaves under .../ab-pairs- refresh/* (0 rejections)	2 -clone-2	1 -clone-2

Total: 27 events (18 substantive + 6 housekeeping + 3 egress-probe).

plan_of_record.md: c56 integrator to register 6 sub-leaves per clone under c33 auto-suffix pattern (slash-separated per Branch B/C convention). Clone-2 already appended 7 rows after the c54 rollup row.

7. Cross-branch disjointness

All writes are disjoint across the three clones:

- **Clone 0:** data/rc10_drums_v2_impl/*, data/recreate_v2/ab_pairs/<sha16>/drums/iter_1/*, scripts/recreate_v2/rc10_drums_v2/*, docs/rc10_drums_v2_*, tests/test_rc10_drums_v2.py, tools/stale/c55_*.py, data/ingestion/egress_status.jsonl (append).
- **Clone 1:** data/rc10_bass_v2_impl/*, data/recreate_v2/ab_pairs/<sha16>/bass/iter_1/*, scripts/recreate_v2/rc10_bass_v2/*, docs/rc10_bass_v2_*, tests/test_rc10_bass_v2.py.
- **Clone 2:** data/rc10_ab_pairs_refresh/*, data/recreate_v2/ab_pairs/<sha16>/{guitar,piano,other_residual,vocals}/iter_1/*, scripts/recreate_v2/rc10_ab_pairs_refresh/*, docs/rc10_ab_pairs_refresh_*, tests/test_rc10_ab_pairs_refresh.py, data/rc10_impl/other_vocals/per_

song/<sha16>/{{vocals,other_residual}}/winner.mid (new files under existing c53 dir; c53 files unchanged).

Zero writes to any READ-ONLY anchor: scripts/palette_render/render_stem.py (214372d9...5b2b); c33/c50/c54 rubrics; c54 v1 data/rc10_drums_bass_impl/*; all three c53/c54 winner_per_stem*.json SHA byte-identical pre==post.

8. Egress state

HTTP 429 + tv_embedded unchanged from c54 baseline across all three probes. Not the two-consecutive media_ok=true unblock signal. Path A per c49 policy.

9. c56 integrator agenda (LINEAR post-merge)

1. **Concat 3 shadow ledgers** into promise_ledger.jsonl (27 rows; per-clone AGENT_FORK_ID routing).
2. **Register 18 sub-leaves** in plan_of_record.md (6 per clone; c33 auto-suffix). Clone-2's 7 rows already landed.
3. **Verify 0-ERROR promise_check** post-registration.
4. **Drums-v2 iteration path:** address the 3 policy calls (WIG cluster-mapping fallback, Dojo G2 window-majority, Disco A near-tie tolerance); propose _infra/rc10-classifier-mapping-fallback-lemma.
5. **Bass-v2 iteration path (v3):** hybridize v1 pyin_mono base + v2 slap/ghost augmentation; re-calibrate slap detector against other-stem bleed; tighten onset delta.
6. **Operator listening loop:** 35 (drums-v2) + 5 (bass-v2) + 40 (refresh) = 80 fresh A/B WAVs ready for operator.
7. **Six-stem gate rollup:** c54 rollup + drums-v2/bass-v2/refresh integration → M_RECREATE_2_LANDS candidacy tally.
8. **c53 clone-2 Chicken Grease chosen-section vs baseline-30s window** — carries forward (fourth branch to touch this now; upstream fix overdue).

10. Session pointers

- Cycle 61 researcher: 32eed67e-9f23-4103-8bc9-ac5fdb0e9a3a
- Cycle 62 worker: 16910231-7cad-480c-8341-0893914bb4fe
- Cycle 63 researcher: 5ea194d7-735e-416b-9b4f-f06b62da3f60

Fanout cadence: LINEAR c49/c50 → FANOUT c51 (3-branch) → LINEAR c52 → FANOUT c53 (RC10 3-branch) → LINEAR c54 (six-stem rollup) → **FANOUT c55 (3-branch: drums-v2, bass-v2, ab-refresh) ← this cycle.** c56 should be LINEAR post-merge integration + v2/v3 iteration decisions on drums/bass.

[END OUTPUT]